Correlation

Correlation is a statistical method to determine whether a linear relationship exists between the variables.

A positive relationship exists when both variables increase or decrease at the same time.

ex: The no of gallons of gas pumped is positively correlated to amount spent on gas.

In a negative relationship, as one variable increases the other variable decreases and vice versa.

ex: Miles travelled is negatively correlated to amount of gas left in your tank.

If there exists no relationship b/w 2 variables then they are said to be non correlated

ex: amount of chocolate someone eats and how many hours they spend on work.

Correlation coefficient (Karl Pearson correlation coefficient) measure to determine the strength of linear relationship b/w 2 variables. It is denoted by r.

The range of r is from -1 to 1. If there is a strong linear relationship b/w variables the value of r will be close to 1. If there is a strong negative linear relationship b/w variables, the value of r will be close to -1. When there is no linear relationship b/w variables or only weak relationship, the value of r will be close to 0.

Formula for the Correlation Coefficient r

$$r = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{\sqrt{[n(\Sigma x^2) - (\Sigma x)^2][n(\Sigma y^2) - (\Sigma y)^2]}}$$

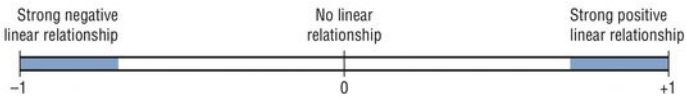
where n is the number of data pairs.

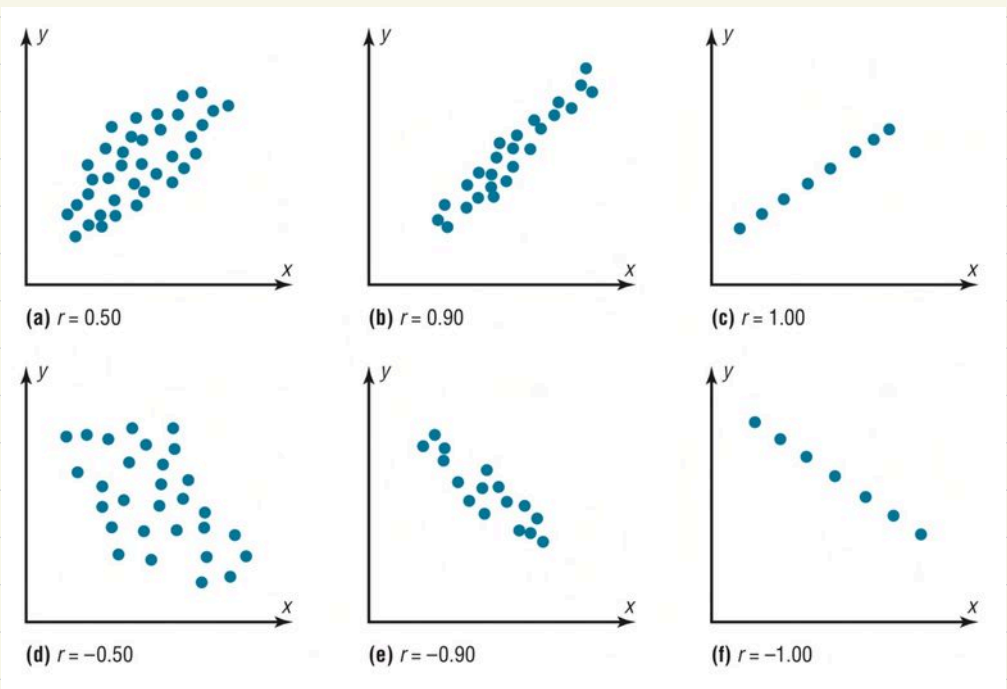
$$r = \frac{\sum (x-\bar{x})(y-\bar{y})}{n\,\sigma_x\,\sigma_y}$$

where

$$\sigma_x = \sqrt{\frac{\sum (x-\bar{x})^2}{n}},$$

$$\sigma_y = \sqrt{\frac{\sum (y-\bar{y})^2}{n}}$$





Alternate formula for r:

$$r = \frac{\sum X_i Y_i}{\sqrt{\sum X_i^2 \sum Y_i^2}}$$

where $X_i = x_i - \bar{x}$, $Y_i = y_i - \bar{y}$

Find the correlation coefficient for following data

x	1	2	3	4	5
y	2	5	3	8	7

Sol: $n = 5$

$$\bar{x} = \frac{1+2+3+4+5}{5} = \frac{15}{5} = 3$$

$$\bar{y} = \frac{2+5+3+8+7}{5} = \frac{25}{5} = 5$$

x_i	$X_i = x_i - \bar{x}$	X_i^2	y_i	$Y_i = y_i - \bar{y}$	Y_i^2	$X_i Y_i$
1	-2	4	2	-3	9	6
2	-1	1	5	0	0	0
3	0	0	3	-2	4	0
4	1	1	8	3	9	3
5	2	4	7	2	4	4
		$\sum X_i^2 = 10$			$\sum Y_i^2 = 26$	$\sum X_i Y_i = 13$

$$\therefore r = \frac{\sum X_i Y_i}{\sqrt{\sum X_i^2 \sum Y_i^2}} = \frac{13}{\sqrt{10 \times 26}} = \frac{13}{\sqrt{260}} = 0.8060$$

Alternate method:

x	y	x^2	y^2	xy
1	2	1	4	2
2	5	4	25	10
3	3	9	9	9
4	8	16	64	32
5	7	25	49	35
$\Sigma x = 15$	$\Sigma y = 25$	55	151	88

$$r = \frac{n\Sigma xy - \Sigma x \Sigma y}{\sqrt{(n\Sigma x^2 - (\Sigma x)^2)(n\Sigma y^2 - (\Sigma y)^2)}} = 0.8060$$

There is a strong positive relationship b/w x & y.

2)

Company	Cars (in ten thousands)	Revenue (in billions)
A	63.0	$7.0
B	29.0	3.9
C	20.8	2.1
D	19.1	2.8
E	13.4	1.4
F	8.5	1.5

Sol.

Company	Cars x (in 10,000s)	Revenue y (in billions)	xy	x^2	y^2
A	63.0	7.0	441.00	3969.00	49.00
B	29.0	3.9	113.10	841.00	15.21
C	20.8	2.1	43.68	432.64	4.41
D	19.1	2.8	53.48	364.81	7.84
E	13.4	1.4	18.76	179.56	1.96
F	8.5	1.5	12.75	72.25	2.25
	$\Sigma x = 153.8$	$\Sigma y = 18.7$	$\Sigma xy = 682.77$	$\Sigma x^2 = 5859.26$	$\Sigma y^2 = 80.67$

Step 3 Substitute in the formula and solve for r:

$$r = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{\sqrt{[n(\Sigma x^2) - (\Sigma x)^2][n(\Sigma y^2) - (\Sigma y)^2]}}$$

$$= \frac{(6)(682.77) - (153.8)(18.7)}{\sqrt{[(6)(5859.26) - (153.8)^2][(6)(80.67) - (18.7)^2]}} = 0.982$$

The correlation coefficient suggests a strong relationship between the number of cars a rental agency has and its annual revenue.

3) Compute the value of the correlation coefficient for the data obtained in the study of the number of absences and the final grade of the seven students in the statistics class given in table.

Student	Number of absences x	Final grade y (%)
A	6	82
B	2	86
C	15	43
D	9	74
E	12	58
F	5	90
G	8	78

Sol :

Student	Number of absences x	Final grade y (%)	xy	x^2	y^2
A	6	82	492	36	6,724
B	2	86	172	4	7,396
C	15	43	645	225	1,849
D	9	74	666	81	5,476
E	12	58	696	144	3,364
F	5	90	450	25	8,100
G	8	78	624	64	6,084
	$\Sigma x = 57$	$\Sigma y = 511$	$\Sigma xy = 3745$	$\Sigma x^2 = 579$	$\Sigma y^2 = 38{,}993$

Step 3 Substitute in the formula and solve for r:

$$r = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{\sqrt{[n(\Sigma x^2) - (\Sigma x)^2][n(\Sigma y^2) - (\Sigma y)^2]}}$$

$$= \frac{(7)(3745) - (57)(511)}{\sqrt{[(7)(579) - (57)^2][(7)(38{,}993) - (511)^2]}} = -0.944$$

The value of r suggests a strong negative relationship between a student's final grade and the number of absences a student has. That is, the more absences a student has, the lower is his or her grade.

4) Compute the value of the correlation coefficient for the data given in Example for the age and wealth of the richest persons in the United States.

$$r = \frac{\sum XY}{\sqrt{\sum x^2 \sum y^2}} = \frac{360}{\sqrt{204 \times 818}} = 0.8812$$

Regression coefficients :

$$b_{yx} = \frac{\sum XY}{\sum X^2} \quad , \quad b_{xy} = \frac{\sum XY}{\sum Y^2}$$

$$b_{yx} = \frac{360}{204} = 1.7647 \quad , \quad b_{xy} = \frac{360}{818} = 0.4401$$

The regression liner of y on x :

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

$$y - 15 = 1.7647(x - 7)$$

$$\therefore y = 1.7647x + 2.6471$$

The regression liner of x on y

$$x - \bar{x} = b_{xy}(y - \bar{y})$$

$$x - 7 = 0.4401(y - 15)$$

$$\therefore x = 0.4401y + 0.3985$$

2) In a partially destroyed lab, record of analysis of correlation data, the following results only are legible. variance of $x = 9$, Regression equations $8x - 10y + 66 = 0$, $40x - 18y = 214$. What are i) mean values of x & y ii) correlation coefficient b/w x & y iii) S.D. of y

Sol:

i) Since both lines pass through $(\bar{x}, \bar{y})$

$$8\bar{x} - 10\bar{y} + 66 = 0,$$
$$40\bar{x} - 18\bar{y} - 214 = 0$$

Solving, $\bar{x} = 13$, $\bar{y} = 17$

Slope close to 1 or >1: y on x

ii) $\sigma_x^2 = 9$, $\sigma_x = 3$

$8x - 10y + 66 = 0$, $40x - 18y = 214$

$y = 0.8x + 6.6$, $x = 0.45y + 5.35$

$b_{yx} = 0.8$, $b_{xy} = 0.45$

$8x = 10y - 66$

$x = \frac{10}{8}y - \frac{66}{8}$

$x = 1.25y - 8.25$

$18y = 40x - 214$

$y = \frac{40}{18}x - \frac{214}{18}$

$= 2.2x - 11.89$

$r = \sqrt{1.25 \times 2.2}$ ✗ 1.65

$r = \sqrt{b_{yx} \times b_{xy}} = \sqrt{0.36} = \pm 0.6$

Since both regression coefficients are +ve, we take $r = 0.6$.

iii) $b_{yx} = r \frac{\sigma_y}{\sigma_x}$

$\therefore \sigma_y = \frac{b_{yx} \times \sigma_x}{r} = \frac{0.8 \times 3}{0.6} = 4$

$\therefore$ S.D. of $y = 4$

3) Find the equation of regression line, of y on x for the data

Student	Number of absences x	Final grade y (%)
A	6	82
B	2	86
C	15	43
D	9	74
E	12	58
F	5	90
G	8	78

Ans: $y = 102.493 - 3.622x$

4) Find the equation of regression line, of y on x for the data

Company	Cars (in ten thousands)	Revenue (in billions)
A	63.0	$7.0
B	29.0	3.9
C	20.8	2.1
D	19.1	2.8
E	13.4	1.4
F	8.5	1.5

Ans: $y = 0.396 + 0.106x$

Spearman's rank correlation coefficient

Definition: The Spearman rank correlation coefficient r_s is a measure of strength of relationship b/w two variables. It is calculated using ranks of each set of data. If both sets of data have same ranks, r_s will be $+1$. If the sets of data are ranked in exactly the opposite way, r_s will be -1. If there is no relationship b/w the rankings, r_s will be 0.

Formula of r_s for non-repeated (distinct) ranks:

$$r_s = 1 - \frac{6 \sum d^2}{n(n^2-1)}$$

where d = difference in ranks

n = number of data pairs or observations.

Formula of r_s for repeated or tied ranks:

$$r_s = 1 - \frac{6 \left\{ \sum d^2 + \frac{m_1(m_1^2-1)}{12} + \frac{m_2(m_2^2-1)}{12} + \dots \right\}}{n(n^2-1)}$$

where m_i is the no of repetitions of ith rank.

1) Calculate the Spearman's rank correlation coefficient from the following data.

Candidates	1	2	3	4	5
Marks in Tamil	75	40	52	65	60
Marks in English	25	42	35	29	33

Sol:

x	Rank of x (R_x)	y	Rank of y (R_y)	$d = R_x - R_y$	d^2
75	1	25	5	-4	16
40	5	42	1	4	16
52	4	35	2	2	4
65	2	29	4	-2	4
60	3	33	3	0	0
					$\sum d^2 = 40$

Here $n = 5$, $\sum d^2 = 40$

$$r_s = 1 - \frac{6 \sum d^2}{n(n^2-1)} = 1 - \frac{6 \times 40}{5(5^2-1)} = 1 - \frac{240}{5(24)} = -1$$

Interpretation: This is a perfect negative rank correlation which indicates that scorings in the subjects totally disagree. Student who is best in Tamil is weakest in English and vice versa.

A researcher wishes to see if there is a relationship between the number of branches a bank has and the total number of deposits (in billions of dollars) the bank receives. A sample of eight regional banks is selected, and the number of branches and the amount of deposits are shown in the table. Is there a significant linear correlation between the number of branches and the amount of the deposits?

Bank	Number of branches	Deposits (in billions)
A	209	$23
B	353	31
C	19	7
D	201	12
E	344	26
F	132	5
G	401	24
H	126	4

Sol:

Bank	Branches	R_x Rank	Deposits	Rank (R_y)	$d = R_x - R_y$	d^2
A	209	4	23	4	0	0
B	353	2	31	1	1	1
C	19	8	7	6	2	4
D	201	5	12	5	0	0
E	344	3	26	2	1	1
F	132	6	5	7	−1	1
G	401	1	24	3	−2	4
H	126	7	4	8	−1	1
						$\Sigma d^2 = 12$

Let R_x be rank of branches,

R_y be rank of deposits.

$$r_s = 1 - \frac{6\Sigma d^2}{n(n^2-1)} = 1 - \frac{(6 \times 12)}{8(8^2-1)} = 1 - \frac{72}{504} = 0.8571$$

Interpretation: There is very strong agreement b/w no of branches and total no of deposits.

Practice problems:

Calculate rank correlation coefficient from the data

1)

x	78	89	97	69	59	79	68
y	125	137	156	112	107	136	124

Ans: $r_s = 0.96$

2)

Hospitals and Nursing Homes Find the Spearman rank correlation coefficient for the following data, which represent the number of hospitals and nursing homes in each of seven randomly selected states.

Hospitals	107	61	202	133	145	117	108
Nursing homes	230	134	704	376	431	538	373

Ans: $r_s = 0.893$

3)

Homework Exercises and Exam Scores A statistics instructor wishes to see whether there is a relationship between the number of homework exercises a student completes and her or his exam score. The data are shown here. Using the Spearman rank correlation coefficient, determine the type of correlation)

Homework problems	63	55	58	87	89	52	46	75	105
Exam score	85	71	75	98	93	63	72	89	100

Ans: $r_s = 0.933$

Repeated ranks

When 2 or more items have equal values it is difficult to give ranks to them. In such cases the items are given average of the ranks they would have received.

1) Compute the rank correlation coefficient for the following data of the marks obtained by 8 students in Commerce and Mathematics.

Marks in Commerce	15	20	28	12	40	60	20	80
Marks in Mathematics	40	30	50	30	20	10	30	60

Sol:

x	R_x	y	R_y	$d = R_x - R_y$	d^2
15	7	40	3	4	16
20	5.5	30	5	0.5	0.25
28	4	50	2	2	4
12	8	30	5	3	9
40	3	20	7	-4	16
60	2	10	8	-6	36
20	5.5	30	5	0.5	0.25
80	1	60	1	0	0
					$\sum d^2 = 81.5$

$$r_s = 1 - 6\left\{\sum d^2 + \frac{m_1(m_1^2-1)}{12} + \frac{m_2(m_2^2-1)}{12} + \ldots\right\} \Big/ n(n^2-1)$$

Here $m_1 = 2, \quad m_2 = 3$

$$\therefore r_s = 1 - \frac{6\left[81.5 + \frac{2}{12}(2^2-1) + \frac{3}{12}(3^2-1)\right]}{8(8^2-1)}$$

$$= 1 - \frac{6[81.5 + 0.5 + 2]}{504} = 1 - \frac{504}{504} = 0$$

Interpretation: Marks in Commerce and Mathematics are not correlated.

2)

Expenditure on advertisement	10	15	14	25	14	14	20	22
Profit	6	25	12	18	25	40	10	7

Sol:

x	R_x	y	R_y	$d = R_x - R_y$	d^2
10	8	6	8	0	0
15	4	25	2.5	1.5	2.25
14	6	12	5	1	1
25	1	18	4	−3	9
14	6	25	2.5	3.5	12.25
14	6	40	1	5	25
20	3	10	6	−3	9
22	2	7	7	−5	25
					$\sum d^2 = 83.50$

$$r_s = 1 - \frac{6\left\{\sum d^2 + \frac{m_1(m_1^2-1)}{12} + \frac{m_2(m_2^2-1)}{12} + \ldots\right\}}{n(n^2-1)}$$

Here $m_1 = 3$, $m_2 = 2$

$$r_s = 1 - \frac{6\left\{83.50 + \frac{3}{12}(3^2-1) + \frac{2}{12}(2^2-1)\right\}}{8(8^2-1)}$$

$$\therefore r_s = 1 - \frac{516}{504} = 1 - 1.024 = -0.024$$

Interpretation: There is a negative association b/w expenditure on advertisement and profit.

Practice problems:

1)

x	40	50	60	60	80	50	70	60
y	80	120	160	170	130	200	210	130

Ans: 0.429 (moderate degree of positive correlation)

2)

x	50	33	40	10	15	15	65	24	15	57
y	12	12	24	6	15	4	20	9	6	18

Ans: 0.733 (strong positive correlation)

3) The table shows the school enrollments (in millions) at all levels of education for males and females from 2000 to 2007. Determine the type of correlation between the number of males and females enrolled in school? *(Source: U.S. Census Bureau)*

Year	Male	Female
2000	35.8	36.4
2001	36.3	36.9
2002	36.8	37.3
2003	37.3	37.6
2004	37.4	38.0
2005	37.4	38.4
2006	37.2	38.0
2007	37.6	38.4

Ans: 0.9167 (very strong positive correlation)

Multiple Regression

This is used when the statistician thinks there are several independendent variables contributing to the variation of dependent variable.

ex: 1) When a sociologist wants to see if the amount of time children spend watching television and playing video games is related to their weight.

2) When a store manager wants to see if the amount spent on advertising and amount of floor space used for display affect the amount of sales of product.

In multiple regression, there are several independent variables and one dependent variable, and the equation is $y' = a + b_1 x_1 + b_2 x_2 + \ldots + b_K x_K$ where $x_1, x_2, \ldots x_K$ are independent variables.

→ or explanatory variable

Formula: If there are 2 independent variables x_1 & x_2 and dependent variable y, then $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$

$\beta_0, \beta_1, \beta_2$ are parameters which are to be determined.

The normal equations are:

$$\Sigma y = n\beta_0 + \beta_1 \Sigma x_1 + \beta_2 \Sigma x_2$$
$$\Sigma x_1 y = \beta_0 \Sigma x_1 + \beta_1 \Sigma x_1^2 + \beta_2 \Sigma x_1 x_2$$
$$\Sigma x_2 y = \beta_0 \Sigma x_2 + \beta_1 \Sigma x_1 x_2 + \beta_2 \Sigma x_2^2.$$

Solving above eqns, we get values of $\beta_0, \beta_1, \beta_2$.

1. A company produces two different items A and B. The data below shows the sale of these items in one day and the profit made by the company on that day.

x_1 *(Sales of item A)*	8	11	9	8	6	10	7
x_2 *(Sales of item B)*	6	4	5	7	1	1	0
Profit (y)	93.26	89.76	60.78	79.34	28.23	75.83	32.74

Fit the best multilinear model that represents the relationship between sales of A and B and the profit.

Solution: The normal equations corresponding to the regression equation $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$ are:

$$n\beta_0 + \beta_1 \sum x_1 + \beta_2 \sum x_2 = \sum y$$
$$\beta_0 \sum x_1 + \beta_1 \sum x_1^2 + \beta_2 \sum x_1 x_2 = \sum y x_1$$
$$\beta_0 \sum x_2 + \beta_1 \sum x_1 x_2 + \beta_2 \sum x_2^2 = \sum y x_2$$

	x_1	x_2	y	y^2	x_1^2	x_2^2	$x_1 * y$	$x_2 * y$	$x_1 * x_2$
	8	6	93.26	8697.43	64	36	746.08	559.56	48
	11	4	89.76	8056.86	121	16	987.36	359.04	44
	9	5	60.78	3694.21	81	25	547.02	303.9	45
	8	7	79.34	6294.84	64	49	634.72	555.38	56
	6	1	28.23	796.933	36	1	169.38	28.23	6
	10	1	75.83	5750.19	100	1	758.3	75.83	10
	7	0	32.74	1071.91	49	0	229.18	0	0
$\sum =$	59	24	459.94	34362.4	515	128	4072.04	1881.94	209

$$7\beta_0 + 59\beta_1 + 24\beta_2 = 459.94$$
$$59\beta_0 + 515\beta_1 + 209\beta_2 = 4072.04$$
$$24\beta_0 + 209\beta_1 + 128\beta_2 = 1881.94$$
$$\beta_0 = -28.5193, \beta_1 = 9.0031, \beta_2 = 5.3496$$

$$=> \quad y = -28.5193 + 9.0031x_1 + 5.3496x_2$$

Practice problems:

1)

A set of experimental runs was made to determine a way of predicting cooking time y at various values of oven width x_1 and flue temperature x_2. The coded data were recorded as follows:

y	6.40	15.05	18.75	30.25	44.85	48.94	51.55	61.50	100.44	111.42
x_1	1.32	2.69	3.56	4.41	5.35	6.20	7.12	8.87	9.80	10.65
x_2	1.15	3.40	4.10	8.75	14.82	15.15	15.32	18.18	35.19	40.40

Estimate the multiple linear regression equation $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$

Ans: $y = 0.4178 + 2.7719 x_1 + 2.0372 x_2$

2) An experiment was conducted to determine if the weight of an animal can be predicted after a given period of time on the basis of the initial weight of the animal and the amount of feed that was eaten. The following data, measured in kilograms, were recorded:

Final weight(y)	95	77	80	100	97	70	50	80	92	84
Initial weight(x_1)	42	33	33	45	39	36	32	41	40	38
Feed weight(x_2)	272	226	259	292	311	183	173	236	230	235

a) Fit the best multilinear model that represents the relationship of the form $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$

b) Predict the final weight of an animal having an initial weight of 35 kilograms that is given 250 kilograms of feed.